

DeepSequence: Hierarchical Attention Forecaster

DS + Softsign + Experts + Context-Aware Mixer

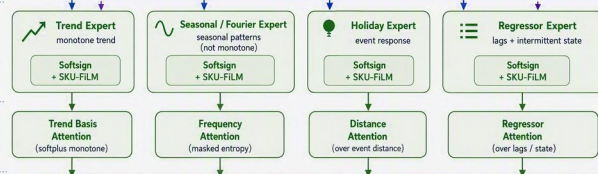
1. INPUTS

Heterogeneous feature groups



2. EXPERTS

Specialized expert networks (different inductive biases)

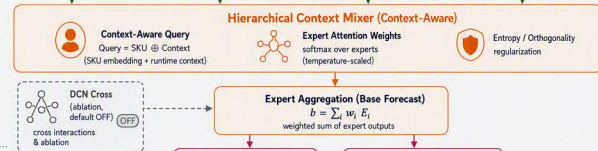


3. INTRA-EXPERT ATTENTION

Capture structure within each expert

4. HIERARCHICAL CONTEXT MIXER

Context-aware mixing across experts



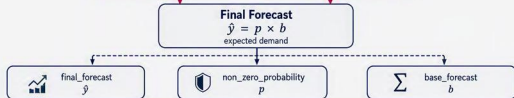
5. FORECAST HEADS

Decompose forecast into magnitude and occurrence



6. OUTPUTS

Final probabilistic forecast



Legend

- Inputs
- Experts & Intra-Expert
- Mixer & Aggregation
- Forecast Heads
- Outputs
- SKU Conditioning (conditioning path)
- Optional / Ablation (default OFF)

Notation

- E_i output of expert i
- w_i attention weight for expert i
- b magnitude (base forecast)
- p non-zero probability
- $\hat{y}$ final forecast

Notes

- Fourier features use fixed frequencies by default; learnable ω optional.
- Seasonal (Fourier) expert is not constrained to be monotonic.
- Regressor expert uses lags and intermittent-state features.
- DCN cross is disabled by default (ablation).

Softsign: bounded activation
SKU-FILM: feature-wise linear modulation by SKU embedding

DeepSequence decomposes demand into occurrence (p) and magnitude (b), combines specialized experts (trend, seasonal/Fourier, holiday, regressor) with hierarchical context-aware attention mixing, and forecasts intermittent and non-intermittent SKUs.